

Rhythm Chawla

Waterloo, ON | +1 (437)-667-5557 | r3chawla@uwaterloo.ca | linkedin.com/in/rhychaw | github.com/rhychaw

EDUCATION

University of Waterloo

Sep 2024 – Apr 2028 (Expected)

Honours Bachelor of Computer Science COOP — AI & HCI, minor Entrepreneurship

Waterloo, ON

– Coursework: Algorithms, Numerical Computation, Probability & Statistics, Linear Algebra, Data Structures.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, Rust, Go, Bash, JavaScript, TypeScript

Systems & Backend: Docker, Kubernetes, GCP, AWS, Django, FastAPI, Spring Boot, Node.js, gRPC, PostgreSQL

WORK EXPERIENCE

Software Engineer Intern (Ongoing)

Jan 2026 – Apr 2026

Carta Maple

Kitchener, ON

- Designed backend services in **Python/Django** for high-volume financial compensation pipelines, processing over **1M records per month** with strict correctness and auditability guarantees.
- Integrated **gRPC** APIs across distributed services, **reducing end-to-end request latency by 30%**.
- Optimized **PostgreSQL** schemas and queries, balancing throughput, consistency, and maintainability; contributed production code under CI pipelines with **unit/integration tests** and **MyPy** static typing.
- Leveraged a toolkit including **Docker, Jenkins, Kubernetes, Datadog, Sentry, and Jira** to ensure reliability, observability, and high system performance.

Associate Conversation Design Intern

May 2025 – Aug 2025

Cresta (Co-op Rating: Excellent)

Remote

- Built an **LLM-powered data auditing system** that processed **10000** conversation logs per run and cut manual QA time by **98%**, enabling reviewers to clear backlogs in hours instead of days.
- Performed QA on AIA bot monitoring, and QA on cresta behavior detection
- Deployed and maintained the system across **12** enterprise environments with varying schemas and traffic patterns, achieving **99%** successful runs over the internship.

Founding Engineer

May 2024 – Aug 2024

G12Uni (Co-op Rating: Outstanding)

Waterloo, ON

- Founded **G12Uni**, an AI platform for university admissions, growing to **40,000** students reached monthly and leading product and ML development for its LLM-powered chatbot, Jado.
- Designed backend microservices supporting **real-time data synchronization** for **2k+** monthly active student users across web and mobile, keeping end-to-end update latency under **200ms** for core flows.
- Built API ingestion layers and structured storage for analytics dashboards, reducing complex reporting queries from minutes to **sub-second** latency via careful schema and index design.
- Deployed containerized services on **GCP** with centralized logging and metrics, improving mean time to detect production issues by **50%** during peak traffic events.

PROJECTS

Clash Royale Analytics Platform

Oct 2025

Python, PostgreSQL, dbt, FastAPI, Streamlit, scikit-learn, XGBoost

- Engineered an end-to-end analytics platform ingesting **50k+** battle logs per day into **PostgreSQL** via batch pipelines and **dbt** transformations with automated data quality checks.
- Trained and deployed ML models (**scikit-learn/XGBoost**) to serve live win-probability predictions, cutting manual matchup analysis time for users by roughly **70%**.

PolyScale — Distributed Code Search & Refactor

Sep 2025

Go, Python, FastAPI, gRPC, Docker

- Prototyped a distributed code search and refactoring system indexing **100k+** source files across large monorepos, exposing a Go HTTP gateway and in-memory search daemon.
- Defined **gRPC** interfaces between Go services and a Python **FastAPI** ML service to deliver embedding-based search and refactor suggestions with **sub-200ms** median response time.
- Enabled local multi-service development with **Docker Compose** and Make-based workflows, cutting environment setup time for new contributors from over an **1 hour** to about **5 minutes**.